Queen's UNIVERSITY

COMPUTING

SCRAP
Satellite Collision Risk Assessment & Prediction

A Two-Stage Machine Learning Framework for Early Collision Risk Estimation in Low Earth Orbit Under Strict Operational Time Constraints

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The Problem: Low Earth Orbit is Crowded

The Kessler Cascade

When two satellites collide at orbital velocities, kinetic energy completely atomizes them — generating thousands of high-speed fragments.

Those fragments tear through other satellites, triggering a chain reaction that could permanently render entire orbital shells unusable.

What we stand to lose permanently:

- GPS & Global Navigation Systems
- Satellite Internet & Communications
- Weather Forecasting Satellites
- International Space Station Access

34,000+

tracked debris objects >10cm in LEO

1 collision

→ 1,000s of new fragments at 28,000 km/h

Cascading Effect

Each collision exponentially increases the debris field — the cascade becomes self-sustaining

Alert Fatigue: When Too Many Warnings = No Warning

99%

of conjunction alerts are
FALSE POSITIVES

The consequence:

- Operators receive hundreds of alerts per week per satellite
- Psychological alert fatigue causes operators to distrust the system
- Critical genuine threats are buried under noise
- Wasted fuel on unnecessary avoidance maneuvers (~$10,000 each)

The real danger isn't a lack of warnings — it's an excess of wrong ones.

Why Physics-Only Methods Break Down

The Chan Formulation

Computes collision probability as a 2D integral over the relative position density, projected onto the collision cross-section plane.

Core assumption:

Satellite position uncertainty follows a perfect Gaussian bell curve

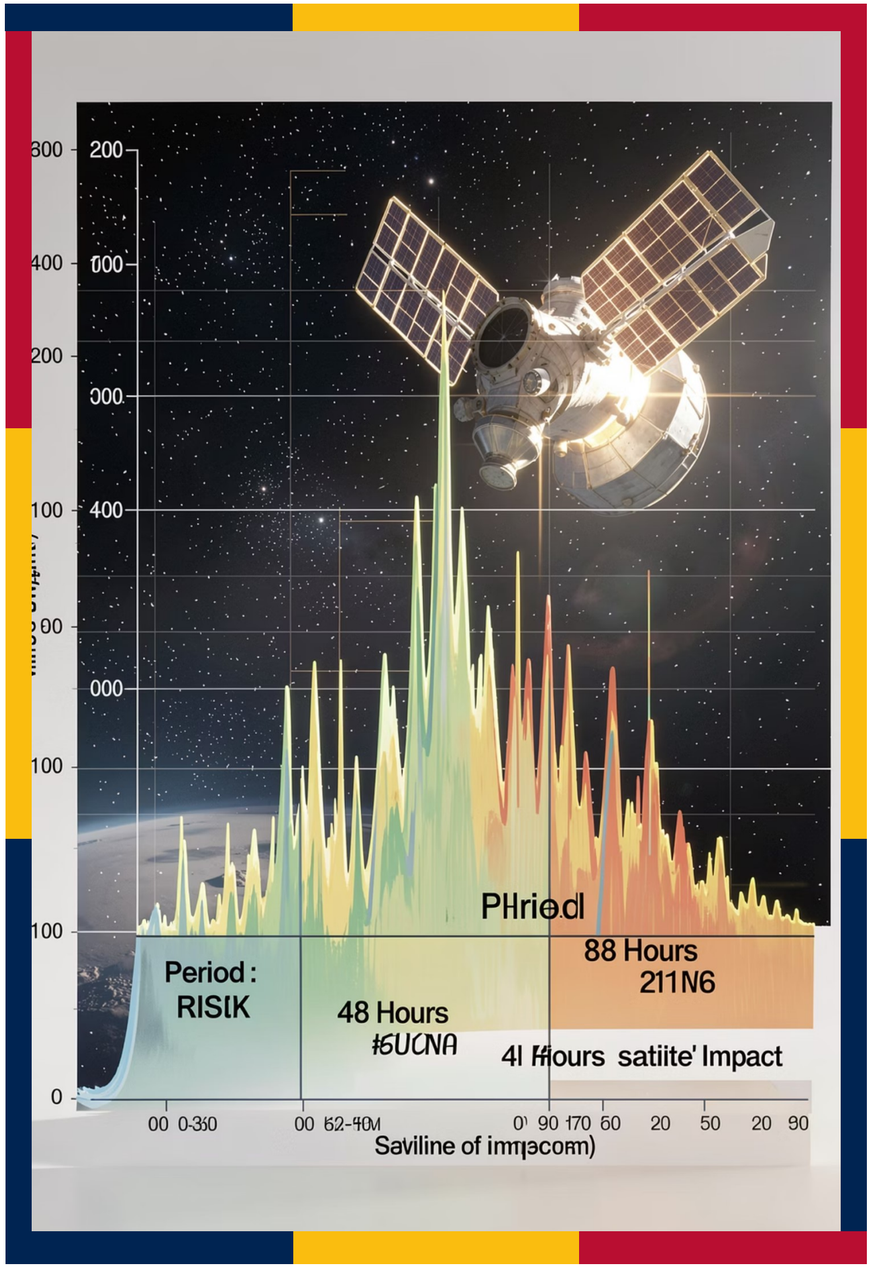
Mathematically rigorous — but physically naive.

When It Fails — Solar Activity

1. Solar radiation heats the upper atmosphere
2. Atmosphere expands → unpredictable drag forces
3. Along-track uncertainty develops non-Gaussian tails
4. Covariance ellipsoids become unreliable
5. Safety perimeters must be massively enlarged
6. Enlarged perimeters overlap constantly → false alarms flood in

The Opportunity:

Physics-based methods cannot leverage statistical patterns hidden in historical CDM time series. Machine learning can !



SCRAP: Our Hypothesis & Dataset

"It is possible to reliably identify high-risk conjunction events — defined as collision probability ≥ 10^{-6} — at least 48 hours before the Time of Closest Approach, using only telemetry data available at that boundary."

162,634
CDM records in total

13,154
unique close-approach events

103
features per record

Feature Categories:

- Kinematics — RTN relative position & velocity
- Covariance matrices — full 3×3 position covariance for both objects
- Space Weather Indices — F10.7 solar flux, AP, SSN

Mining 162,000 Telemetry States for the 1% Threat

Data Source & Scale

162,634
individual CDM records

13,154
unique close-approach events

103
numerical features

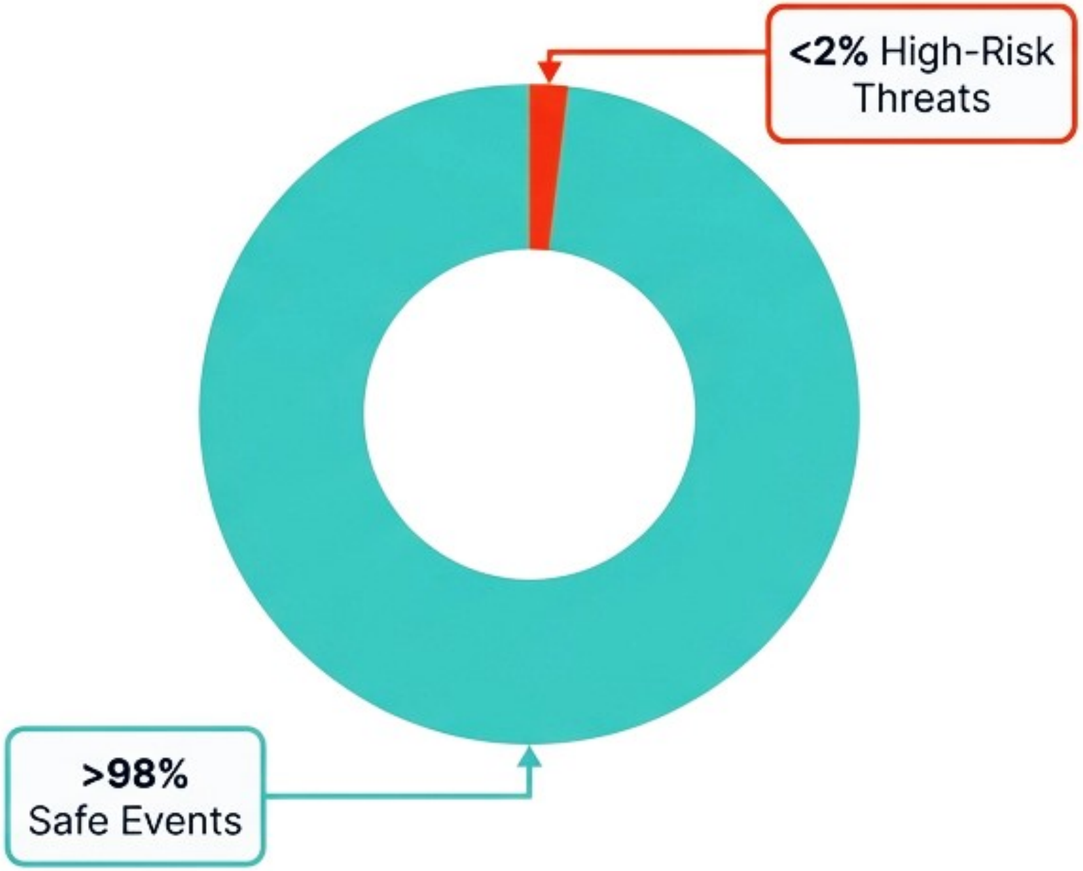
Telemetry Inputs

Kinematics: relative position/velocity

Uncertainty Matrices: 3D radar error

Space Weather: F10.7, AP index

The Challenge: Extreme Imbalance



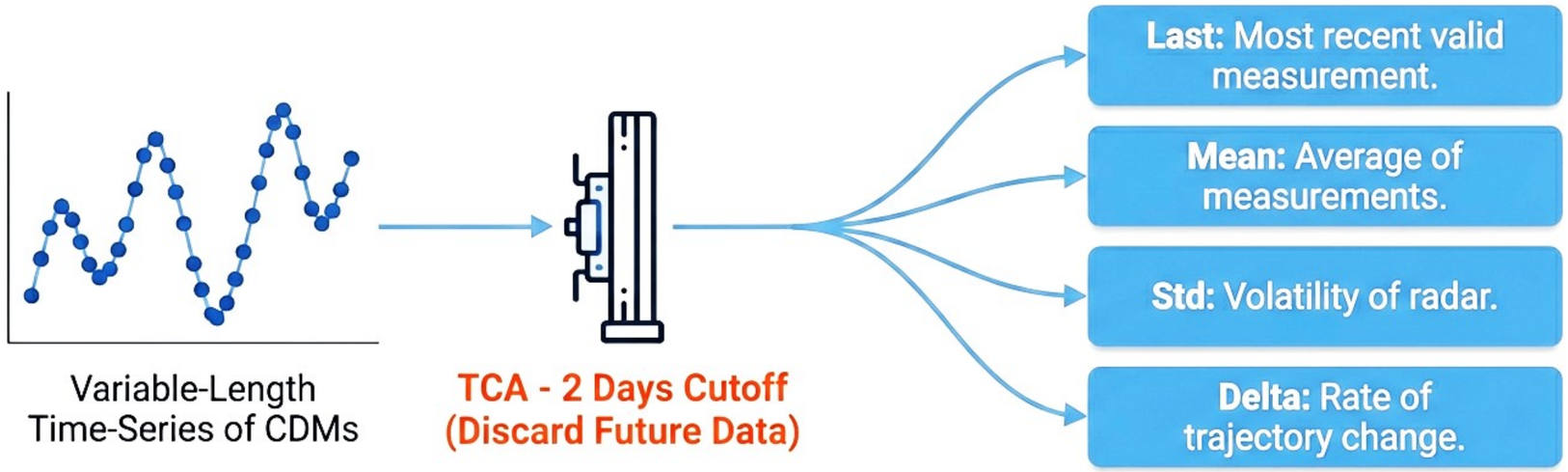
Time-Series Flattening → 1,208 Features

Each event has variable-length CDM sequences. We apply 11 statistical aggregations per feature to produce a fixed-length tabular vector:

last Most recent observation	**mean** Central tendency	**std** Variability over time	**min / max** Observable extremes	**delta** Total drift	**slope** Rate of change	**last2_change** Boundary step anomaly	**change_ratio** Relative spike magnitude

recent_vs_early Medium-term drift	**max_single_jump** Largest single-step spike	★ **Momentum Statistics** Added to detect jump-prone events approaching the danger threshold

103 base features × 11 aggregations + derived physics features(75) = **1,208 features / event**



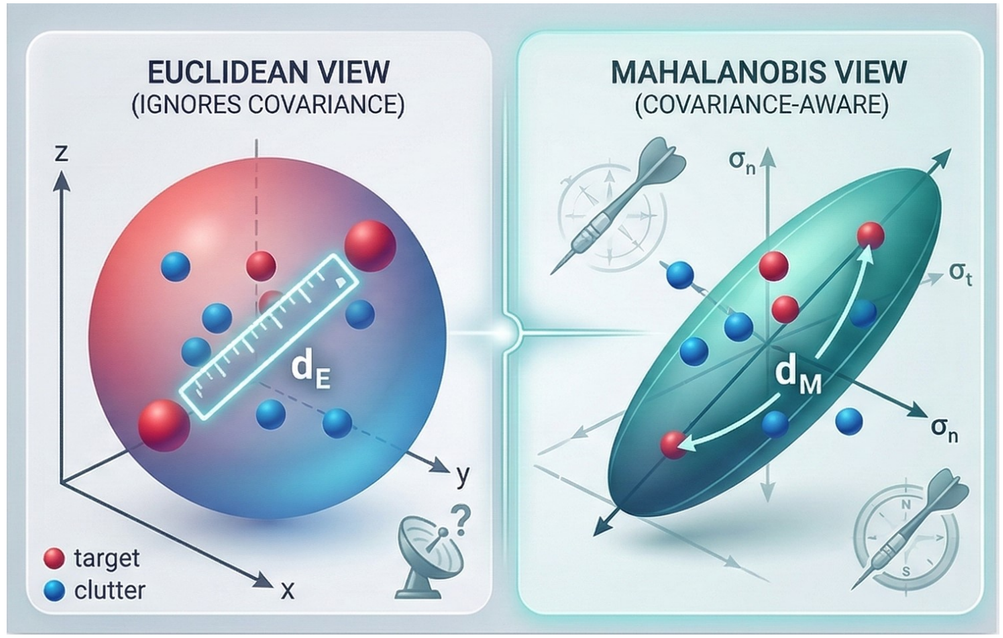
Mahalanobis Distance: Why It Matters

$$d_M = \sqrt{[(\Delta r/\sigma_r)^2 + (\Delta t/\sigma_t)^2 + (\Delta n/\sigma_n)^2]}$$

Covariance-normalized spatial separation

Euclidean Distance

- Measures raw physical distance only
- Treats all axes as equally certain
- Ignores radar tracking quality
- Two objects 10 km apart — but is that measurement reliable?



Mahalanobis Distance ✓

- Scales the gap by the covariance matrix
- Encodes both proximity AND measurement confidence
- A dart 1 cm from bullseye with shaking hand > steady hand
- Physically meaningful: How close are they, really?

Physically interpretable · Covariance-aware · Most impactful single feature in SHAP analysis

Two-Stage Sentinel-Specialist Architecture

The ESA compound loss couples F_2 (classification) with MSE (regression). These are mathematically distinct objectives — we decompose them.

ALL EVENTS

11,942 training events

STAGE 1 — SENTINEL

LightGBM Classifier

- Flags: will this event exceed 10^{-6}?
- is_unbalance = True (auto scale_pos_weight)
- Threshold tuned on OOF → t* = 0.6680
- OOF F_2 = 0.9424

High-risk only

STAGE 2 — SPECIALIST

XGBoost Regressor

- Trained ONLY on 1,293 high-risk events
- Predicts exact $\log_{10}$(risk) magnitude
- Sample weight W = 32.94 (imbalance × β^2)
- RMSE = 0.2246 $\log_{10}$ units

Borderline Promotion Module:

Events in [t*/2, t*) with uncertainty volume above 75th percentile are promoted to −5.99, capturing jump-regime victims likely to spike in the final 48 hours.

Log_{10} Transformation: Taming 12 Orders of Magnitude

$$y = \log_{10}(r)$$

−12	−6	−4	−2	−0.3
$\log_{10}$	$\log_{10}$	$\log_{10}$	$\log_{10}$	$\log_{10}$
Negligible	**Threshold**	**Low-High**	**High**	**Critical**
0.000000000001	0.000001	0.0001	0.01	0.5

- Without transformation: model struggles with numbers spanning 10^{-12} to 10^{-1} — 12 orders of magnitude
- With $\log_{10}$: probabilities become uniformly spaced values from −12 to −1 — numerically stable regression
- MSE_HR is evaluated in probability space (converted back via 10^y) to give the official ESA loss value

Stratified Group K-Fold: Preventing Time-Series Leakage

Brilliant architecture cannot save a flawed validation strategy. Time-series leakage is the silent killer of ML models.

❌ Random K-Fold Split

- CDMs from the same event split across train/test
- Model sees fragments of the same trajectory in training
- Learns to memorize paths, not predict physics
- OOF estimates are artificially inflated — model appears better than it is

✓ StratifiedGroupKFold (groups = event_id)

- Splits strictly by unique event IDs
- No CDM from an event ever appears in both folds
- Stratified: maintains class balance across folds
- Guarantees model is tested only on futures it has genuinely never seen

Training: 11,942 events

1,293 high-risk (10.83%)

Test: 2,167 events

334 high-risk (15.41%)

LRP Baseline: Proving the Task is Non-Trivial

Before testing SCRAP, we evaluated the simplest possible strategy:

Latest Risk Prediction (LRP):

Use the last known risk value at the 2-day boundary as the final prediction. No modelling required.

$$L = \infty$$

Competition Loss

Why infinity?

- At the 2-day boundary, EVERY event's observed risk is below 10^{-6}
- LRP classifies everything as safe — 0 high-risk events flagged
- $F_2 = 0 \rightarrow L = \text{MSE} / F_2 = \text{anything} / 0 = \infty$
- This is the most important diagnostic in the project

Any model beating $L = \infty$ has learned genuine predictive structure from 48-hour-old telemetry.

SCRAP: Final Test-Set Performance

334 true high-risk events in the test set

False Negative — Missed Collision

vs

False Positive — Wasted Maneuver Fuel

326 / 334	97.60%	0.9482	0.2246
High-risk events correctly flagged (TP)	Recall — only 8 missed	F_2 Score	Specialist RMSE ($\log_{10}$ units)

Confusion Matrix

	Predicted Safe	Predicted Danger
True Safe	1,776 (TN)	57 (FP)
True Danger	8 (FN) ★	326 (TP)

Additional Metrics

Precision:	0.8512
Sentinel Separation:	30.4×
MSE_HR (prob space):	2.63×10^{-8}
Official Loss L:	2.77×10^{-8}

ESA Competition Winner: F_2 = 0.733–0.765 → SCRAP F_2 = 0.9482 | Evaluated on our stratified split (note: different test partition from ESA curated set)

Limitations: Intellectual Honesty

01

Solar Cycle Temporal Drift

The sun operates on an 11-year cycle. A model trained on one phase of that cycle will experience predictive drift as atmospheric drag physics shift. The F10.7 solar flux features capture current state but cannot predict cycle transitions. Continuous retraining on recent CDMs is required for operational deployment.

02

Irreducible Bayes Error Floor

Our 2.40% miss rate — 8 events — likely represents conjunction events that undergo covariance updates in the final 48 hours that no system can predict from older telemetry. The laws of orbital dynamics impose a genuine uncertainty floor that cannot be overcome by any algorithm.

03

Evaluation Partition Difference

Our stratified split yields 15.41% high-risk prevalence in the test set vs. the ESA curated partition. F_2 comparisons with competition results should account for this prevalence difference — direct numeric comparison is indicative, not exact.

04

Probability-Space Loss Interpretation

MSE_HR is computed in probability space, not $\log_{10}$ space. High-risk probabilities in $[10^{-5}, 10^{-1}]$ produce squared differences of order $10^{-10} – 10^{-8}$, making $L \approx 0$. The operationally meaningful metrics are F_2, Recall, and Specialist RMSE.

CONCLUSION

SCRAP proves AI can see what physics alone cannot.

1. 1,208-feature physics-informed pipeline effectively captures CDM time-series dynamics including jump-regime events.
2. Decomposing the ESA compound loss into Sentinel + Specialist outperforms any single-model approach.
3. Sample weights calibrated from imbalance ratio × β^2 = 32.94 ground training in the ESA recall priority.
4. F_2 = 0.9482, 97.60% Recall — 326 of 334 high-risk events flagged with only 8 false negatives.
5. 97.68% recall on the hardest jump-regime cases confirms the Borderline Promotion Module works.

A 2.40% miss rate on high-risk events provides sufficient lead time for avoidance maneuvers — eliminating the alert fatigue that makes current physics-only systems unreliable.